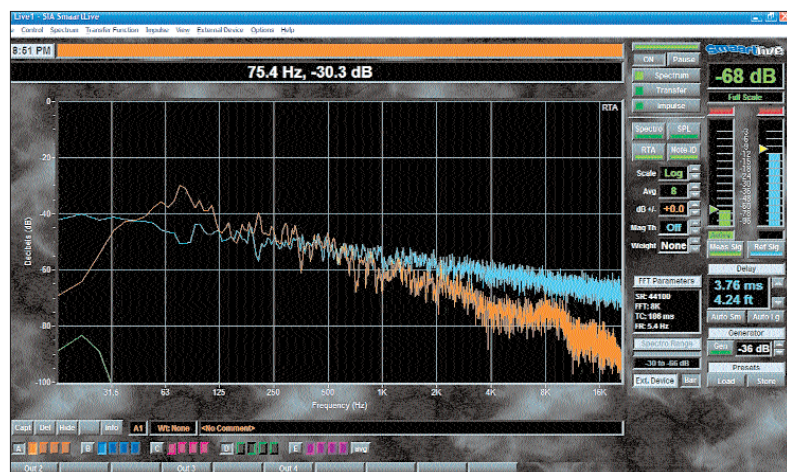




The analysis of the Altec Lansing ATP3 shows that they are pretty bass heavy, and lose out a little in terms of highs. This can be seen in the graph between the 4 to 16 KHz mark

woofer and slim satellites with stands that can also be used as brackets for wall-mounting.

A remote control lets you control the speaker volume and other parameters without the necessity of leaving your seat. The Artis S111R, the Creative duo, as well as the Xfree duo came with wired remotes. The Altec Lansing FX4021 provided both a wired and a wireless remote. Only the Artis S111/FM sported FM radio.

Build Quality

The Altec Lansing FX4021 had a better build than the rest, but was very closely followed by its sibling, the ATP3, and the Logitech X-230. The Zebronics ZEB-SW8000 was decently put together and was the heaviest of the sets, but its knobs weren't the sturdiest. Most of the subwoofers were ported to allow for maximum displacement of air while producing bass. How much of this design translated into actual performance was verified when we tested the performance.



Altec Lansing ATP3
Crystal-clear!

Performance Music

The Logitech speaker sets were found to be the best performers with music of every genre. Not only were the highs crisp and clear, the lows were rich, and the mid-range pleasant. The Altec Lansing ATP3 and FX4021 fiercely contested for second place. While the ATP3 faltered a bit with rock, with flawed bass reproduction, it did produce better *Bhangra* beats and *Ektara* sounds than the FX4021. The performance of the Artis S444 and the S800 was excellent, yet they come at low prices. Lesser-known brands such as Xfree, Tech-Com, and Zebronics exhibited below-average performance, as did the Creatives. The Artis S-100 was the lowest-performing speaker set with low scores in almost every category.

DVD Movies

While none of the speaker sets could flawlessly render the concussion gun boom in *Minority Report*, the Logitech X-230 and the Altec Lansing

How We Tested

Our test PC comprised an AMD Athlon64 3800+ processor running at a stock speed of 2.0 GHz, a WinFast 6150K8MA motherboard, 1 GB of 400 MHz Corsair XMS DDR RAM, a Seagate Barracuda 120 GB SATA150 hard drive, and the Creative Sound Blaster X-Fi Elite Pro sound card, running Windows XP Professional with SP 2. The latest drivers and DirectX version were installed.

We put the speakers through an exhaustive set of tests, including three DVD movie tests, multiple music and sound tests, as well as a game test.

The game test was the *Half-Life 2* Demo "d3_c17_12". We ran this demo with sound settings at "high," while the visual quality was set to "low" to reduce CPU overhead. The sound was configured according to the number of channels available on the speaker sets. This demo blends voice commands with firing and explosions, and is good for testing the capabilities of speakers when reproducing varying sounds. A speaker set that reproduces the faintest of directional sounds with clarity, such as footsteps, is good for gaming and can save you from getting fraggled.

A DTS sampler disc with a couple of DVD movie clips and a 5.1 soundtrack were used as the DVD test material. This sampler disc contains high-quality content and is bundled with high-end Creative sound cards—perfect for stressing out speakers. We noted how clearly the speaker sets managed to produce faint footsteps, the

twang sound of the soldier stepping on the trip wire, and explosion bass in *Behind Enemy Lines*. In *The Fast and The Furious*, we noted how well they reproduced the sound of breaking glass as well as the humming and roaring of the engines. Finally, we played *Minority Report* and listened to the blast of extremely low bass produced by the Concussion rifle.

We listened to Eric Clapton's *Broken* to test DVD Audio. In this song, there are certain details such as a "blow" on the microphone at the start, which only a few speaker sets managed to reproduce correctly. There is the guitar twang as well as the squeaky sound of fingers moving along the strings, which we noted. The soundtrack was rated for bass, treble and vocals.

For music playback, we divided the tests into Western and Indian. This was further sub-divided into different genres. In all the cases, we looked for clarity and for distortion.

The Western music playlist included the following.

Rock *With Or Without You* by U2. This song has some deep bass as well as clear vocals; we noted these.

Jazz We chose a couple of Dave Brubeck numbers. Faithful reproduction of the sax and piano were noted in this case. Drums should sound as they would in the background, as they were positioned that way during the recording.